Seneca | SCHOOL OF INFORMATION TECHNOLOGY ADMINISTRATION & SECURITY

Project 2: Transactional Machine Learning (TML) Solution Studio (SS) and Cisco Packer Tracer(CPT)

Cyberthreat Detection and Real-Time Network Monitoring with Kafka Cloud, Python, PrivateGPT, Docker, MariaDB, TMUX, HiveMQ, MQTT, and Real-Time Dashboard Running in Linux

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Introduction

In today's increasingly connected world, the need for real-time monitoring and analytics of IoT devices, networks, and security systems has never been more critical. The rapid growth of connected devices and the vast amounts of data they generate present unique challenges for businesses looking to optimize their operations, ensure security, and maintain efficiency. Traditional data processing methods, which often rely on batch processing or SQL databases, cannot keep up with the scale and speed required for real-time insights. As a result, organizations are left with delayed responses to failures, inefficiencies, and potential security breaches that could have been prevented with timely intervention.

In this document, we are presenting a Transactional Machine Learning Solution Studio (TSS) project, that is specifically designed to address these challenges. By integrating real-time data processing, machine learning, and interactive visualization, TSS enables organizations to stay ahead of issues before they escalate. Whether it is predicting device failures in an IoT ecosystem, detecting network anomalies, or preventing security breaches, TSS provides businesses with the insights they need to make immediate, data-driven decisions. With the ability to scale to meet the needs of growing organizations and adapt to various industries, this solution is a powerful tool for improving operational efficiency and strengthening security in a data-driven world.

What is Transactional Machine Learning Solution Studio – TSS?

The Transactional Machine Learning Solution Studio is a powerful tool that helps businesses manage and analyze data from their IoT devices and networks in real-time. It uses advanced technology to quickly process large amounts of data and detect issues like device failures or network problems before they become big problems. By using machine learning, TML can predict potential failures and spot anomalies, helping businesses stay ahead of any issues. The solution processes data at the individual device level, providing accurate, real-time insights tailored to each device's needs.

TSS is easy to scale and deploy, thanks to its use of Docker containers and Kubernetes, making it flexible enough to grow with your business. It also includes an interactive dashboard that displays all the data and insights in real-time, so you can quickly understand the health of your devices and networks. Whether you're looking to predict when equipment might fail or ensure your network stays secure, TML helps businesses stay proactive and efficient by making data-driven decisions faster.

Problem Addressed

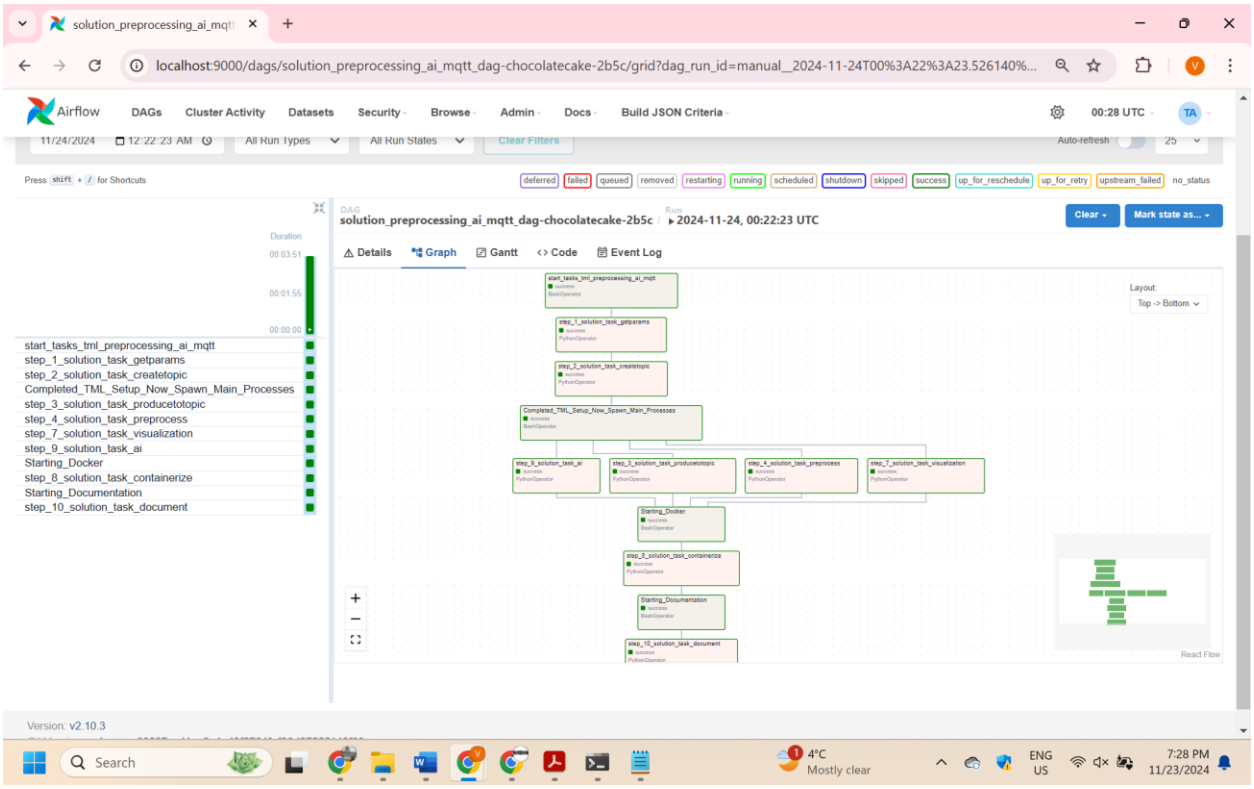
Organizations today face significant challenges in processing and analyzing the vast amounts of data generated by IoT devices, networks, and security systems. Traditional systems often struggle with the speed and scalability required for modern, high-volume data streams. This limitation leads to inefficiencies, missed opportunities, and, in some cases, costly downtime or security breaches.

The solution addresses these challenges by providing a robust, scalable platform that processes real-time data streams, detects anomalies or device failures before they escalate, and offers immediate insights into network health and security. It improves operational efficiency, strengthens security, and minimizes the risk of unforeseen disruptions.

How TSS works?

TSS combines real-time data processing with machine learning to handle vast amounts of data generated by IoT devices and networks. It is designed to provide insights and predictions that help businesses make informed, proactive decisions.

Solution Process Diagram



This is the Solution Graph where we can see the process of the

1. **Real-Time Data Processing:**

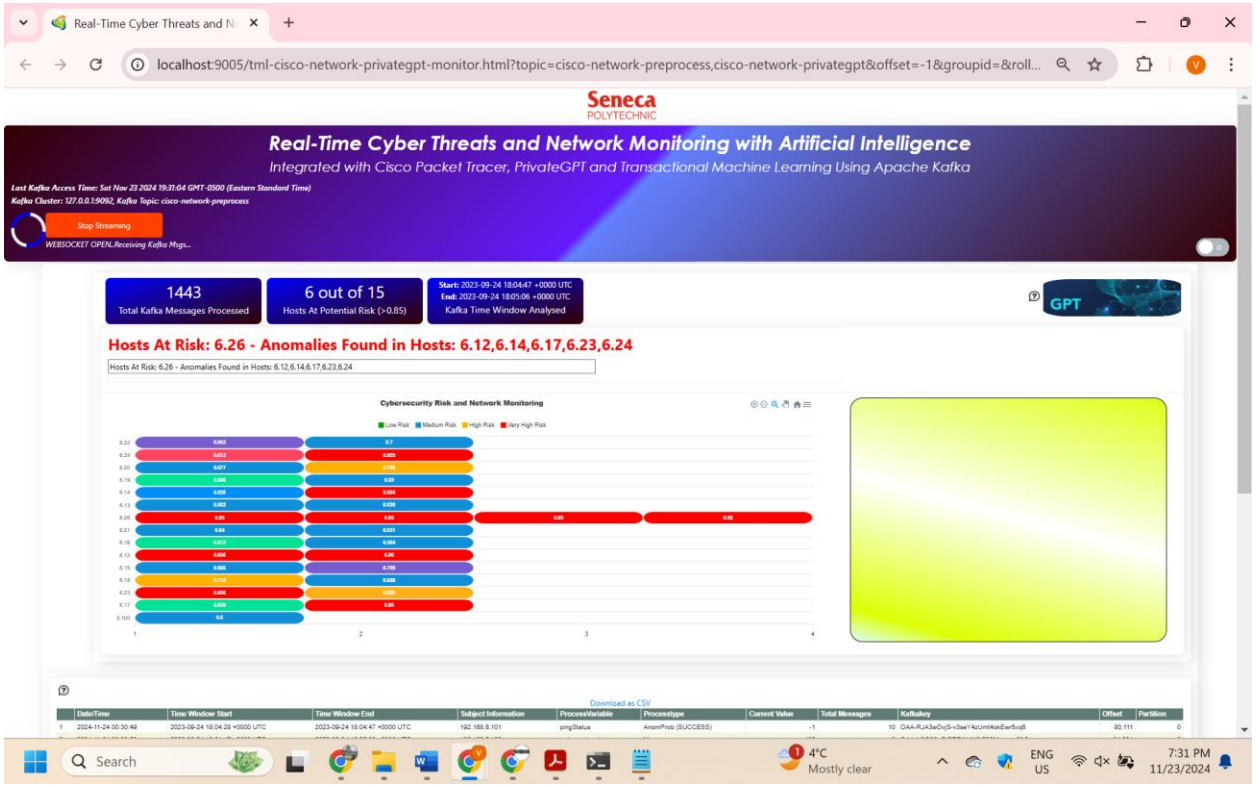
- TML uses **JSON paths** to extract necessary fields from incoming JSON data streams. These fields are grouped and processed using a user-defined **jsoncriteria**, allowing users to specify which data points are of interest (e.g., device serial numbers, timestamps, location, values)
- TML performs in-memory processing of data streams, allowing data to be analyzed as it arrives. This eliminates delays caused by traditional batch processing or SQL queries and ensures immediate insights, making it ideal for environments with large volumes of data.

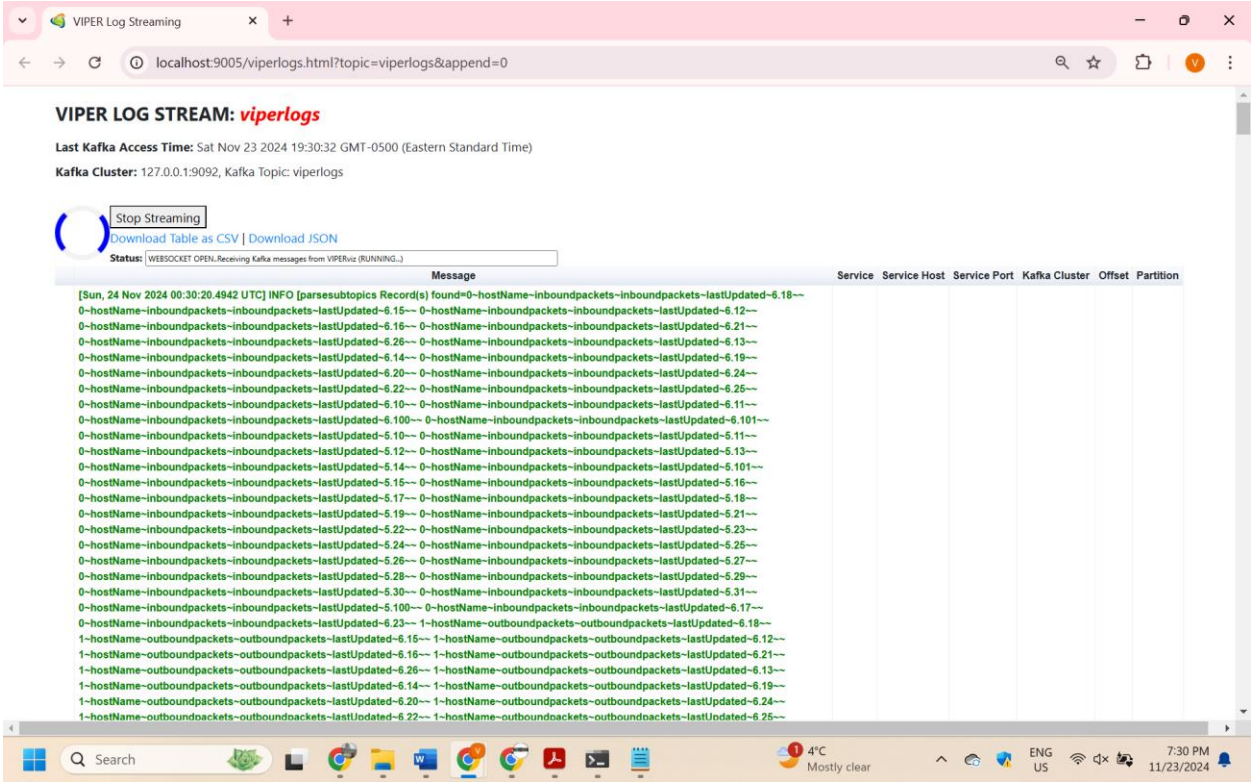
- It operates at the entity level, meaning each device's data is processed individually. This approach leads to more accurate and actionable insights, tailored to the unique characteristics of each device or network component.

2. **Data dashboarding:**

   TML leverages Viperviz, a visualization tool that uses WebSockets to display processed data streams on a dashboard. The dashboard provides:

- Real-time visualization: As data flows through the system, it is dynamically visualized on the dashboard, allowing users to monitor device health and network status.
- Alerting and Anomaly Indicators: Devices are shown with color-coded indicators (e.g., red for failure risks, green for normal operation), based on predefined anomaly detection models and thresholds.





3. Data storage in confluent Kafka Cloud

Data is stored in Kafka Cloud as Kafka topics. These topics are logical channels where different types of data streams (such as IoT sensor readings, event logs, etc.) are categorized. TML processes these topics in real-time by:

- Ingesting data from Kafka topics.
- Performing analysis and processing in-memory.
- Sending results back to Kafka or directly to the dashboard for visualization.

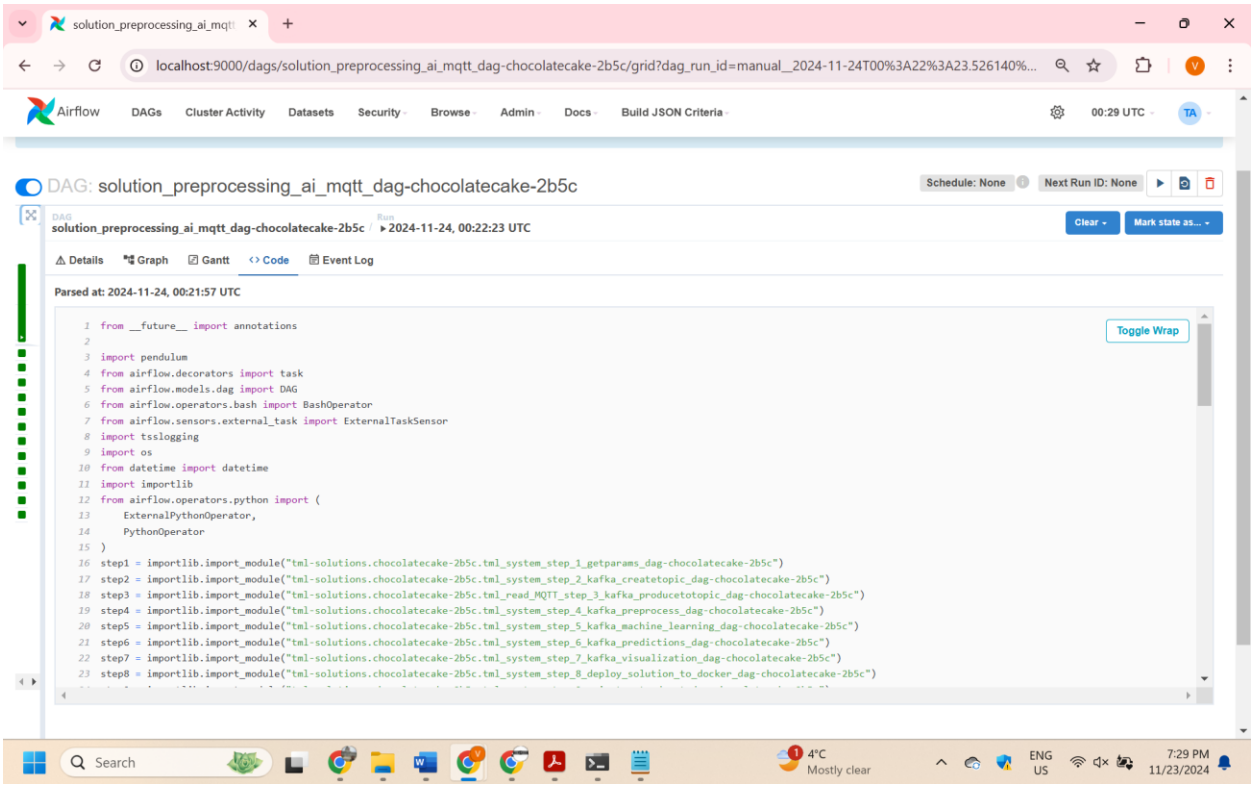
Kafka is highly scalable, handling billions of messages per second, and allows TML to scale accordingly with the growth of data volume and devices.

4. Cisco Packet Tracer

We are using Cisco Packet Tracer to simulate network activity in real-time, allowing for the creation of scenarios such as device failures, network anomalies, or security breaches. This simulated data is sent to Kafka, where TML processes it for:

- Anomaly detection: Detecting unusual network traffic patterns or unauthorized access.
- Device status monitoring: Monitoring the health of virtual devices and identifying any issues before they lead to network downtime.
- Real-time response: TML can trigger actions based on detected anomalies, such as alerting network administrators or taking corrective actions.

Solution code

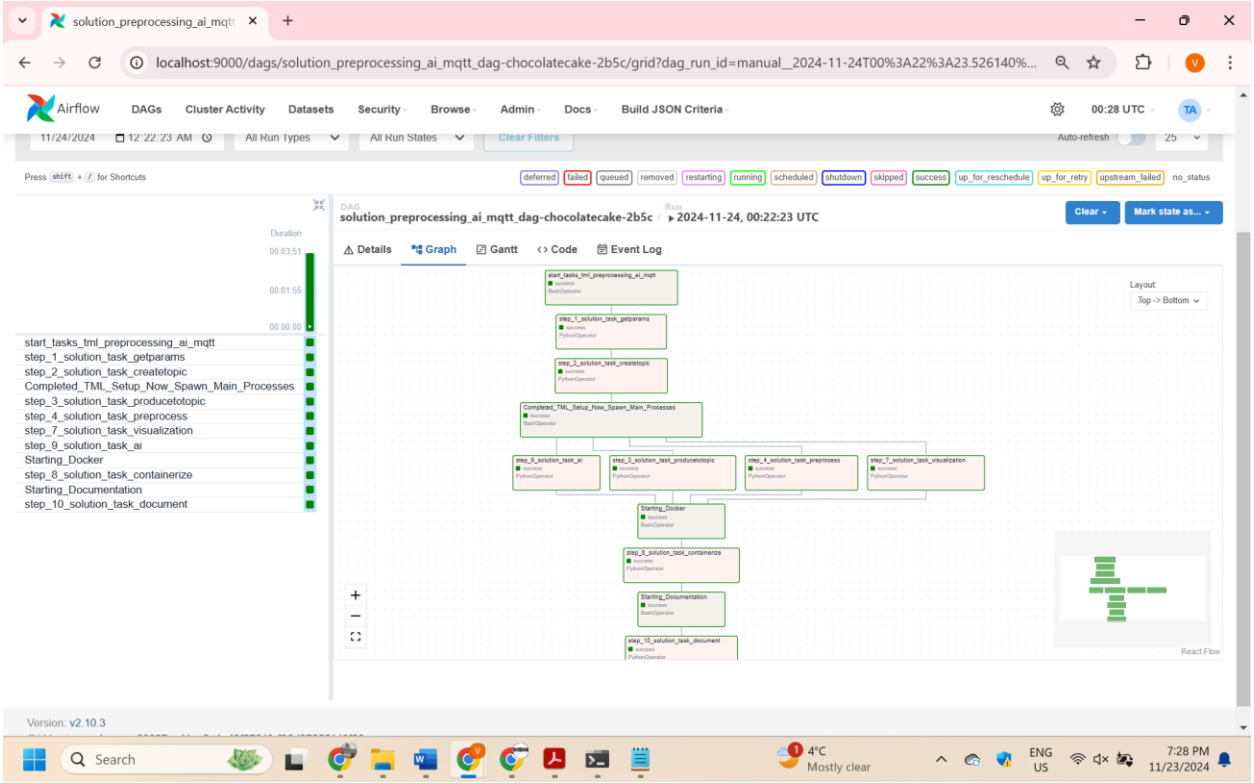


- Link to Docker container https://hub.docker.com/r/vanesalcb/tssproject2-2b5c-amd64
  docker pull vanesalcb/tssproject2-2b5c-amd64
- Link to Github https://github.com/vanesalcb06/tssproject2-2b5c.git

  https://github.com/vanesalcb06/raspberrypi/tree/main/tml-airflow/dags/tml-solutions/tssproject2-2b5c
- Link to the documentation generated from TSS 7. https://tssproject2-2b5c.readthedocs.io/en/latest/

Core Technology Components

Several advanced technologies work together to deliver real-time data processing, machine learning, and visualization:



- Apache Kafka: Acts as the central messaging system for ingesting and distributing data streams. Kafka topics manage real-time data flow and ensure that the system can scale to handle billions of data points.
- MAADS-VIPER: The engine responsible for in-memory processing. It handles the data processing tasks, including anomaly detection, and pushes results back to Kafka for further analysis or visualization.
- MAADS-HPDE (AutoML Engine): Automates the process of building, training, and tuning machine learning models in real-time. By continuously analyzing the data, it predicts potential failures or detects anomalies, empowering businesses to take preemptive action.
- Viperviz: A real-time visualization tool that displays processed data on an interactive dashboard. The dashboard helps users monitor device health and network status, providing real-time alerts on anomalies or failures.

- Docker & Kubernetes: The entire solution is containerized using Docker, making it easy to deploy and manage. Kubernetes orchestrates the containers, ensuring scalability and high availability. The system can scale automatically based on the volume of data, handling increasing numbers of devices and data points without compromising performance.

Integration & Data Flow

The system's components are seamlessly integrated to work together as follows:

- Data Ingestion: IoT devices, network components, or even simulated data from tools like Cisco Packet Tracer send data to Kafka topics. These topics serve as channels for the real-time data flow.
- Processing: The data is processed by MAADS-VIPER, which performs the necessary analysis in-memory, detecting anomalies, trends, and potential issues as they arise.
- Machine Learning: MAADS-HPDE uses the data from Kafka to build machine learning models that predict device failures or detect network threats. These models are continuously updated as new data is ingested.
- Visualization: Processed data is sent to the Viperviz dashboard, where it is displayed in real-time. Users can track device health, performance trends, and receive alerts about any detected anomalies.

The Importance of Artificial Intelligence and Machine Learning

Artificial intelligence (AI) and machine learning (ML) are fundamental to the solution's ability to deliver actionable insights and improve operational efficiency. Their importance lies in several key areas:

- Real-Time Decision Making: AI and machine learning help systems analyze large amounts of data quickly. They spot patterns, unusual activity, and trends faster than people or traditional systems can. This speed is vital in fields that need quick responses, like cybersecurity, healthcare, and manufacturing..
- Predictive Analytics: Machine learning models learn from historical and real-time data, enabling the prediction of future events such as equipment failures, network threats, or shifts in consumer demand. This predictive capability allows organizations to proactively address issues before they escalate, saving costs and preventing disruptions.

Deployment & Scalability

The solution is deployed in Docker containers, which ensures portability and makes the deployment process efficient across any infrastructure—whether on-premises or in the cloud. Kubernetes manages the containers, automatically scaling the solution as the number of devices or volume of data grows. This makes the system highly flexible, and capable of handling everything from small deployments to large, enterprise-level systems.

Comparison to Other Solutions

What differentiates this solution is its ability to perform real-time, in-memory data processing at the entity level, rather than relying on batch processing or centralized databases. This approach provides faster, more accurate results and allows for immediate responses to potential issues.

Additionally, the system incorporates AutoML for predictive maintenance and anomaly detection, which sets it apart from traditional solutions that may not offer these advanced capabilities. By visualizing data in real-time, businesses gain actionable insights that they can use to improve operations and prevent problems before they escalate.

Why is This Solution Important?

In industries like manufacturing, healthcare, and cybersecurity, being able to detect and address issues before they become major problems is crucial. The solution helps organizations improve operational efficiency, reduce downtime, and enhance security by providing real-time, actionable insights. Whether it's predicting a failure in industrial equipment or identifying a potential security breach in a network, the solution ensures that businesses can act quickly and effectively.

Extension to Other Industries

While the solution was designed for IoT and network monitoring, its flexibility means it can be extended to a wide range of industries:

- Manufacturing: It can be used for predictive maintenance of machinery, helping manufacturers reduce unplanned downtime and extend the life of equipment.
- Healthcare: Continuous monitoring of patient vitals and predicting potential health issues before they occur can save lives and improve care delivery.
- Retail: Real-time inventory tracking and asset management can optimize the supply chain, ensuring businesses are always prepared to meet demand.
- Finance and Banking: Fraud detection can be enhanced through real-time analysis of transaction data to identify unusual patterns or activities. The system can also help optimize financial decision-making by analyzing market trends.

The system's ability to adapt to different industries ensures that it can add value across various business sectors.

Scaling the Solution

As businesses grow, the volume of data they generate also increases. This solution is designed to scale seamlessly with that growth. By leveraging Kubernetes for container orchestration and Kafka for data stream management, the system can expand as needed, without sacrificing performance.

The solution is also cloud-native, which means it can be deployed in the cloud for even greater scalability, availability, and flexibility. Businesses can take advantage of hybrid or multi-cloud environments to ensure that their system is always up and running, no matter how large or complex their operations become.

Concluding Thoughts

The solution offers a comprehensive, scalable platform for real-time monitoring, predictive analytics, and anomaly detection. By integrating in-memory data processing, AutoML, and real-time visualization, the system empowers businesses to improve operational efficiency, enhance security, and stay ahead of potential issues. The solution's adaptability makes it a valuable asset for any organization looking to leverage real-time data for smarter decision-making and improved business outcomes.

References

Maurice, S., & Ghaderi, K. (2024). *Transactional Machine Learning (TML)*. Amazon. Available at: https://a.co/d/4kYJ1Cb

TML Documentation. (2024). *Transactional Machine Learning (TML) build documentation*. https://tml.readthedocs.io/en/latest/tmlbuilds.html